AI SKIN INTELLIGENCE AND PERSONALIZED SKINCARE PLANNER

Project Report

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A web-based platform for skin assessment, lifestyle monitoring, personalized skincare routines, recommendations, and consultation booking.

Academic Project Report

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1. Abstract

AI Skin Intelligence and Personalized Skincare Planner is a web-based project designed to organize basic skincare information and provide personalized guidance using information supplied by users. The platform brings together skin profiling, skin assessment, lifestyle tracking, hydration and sleep tracking, skincare routine generation, ingredient information, product recommendations, progress tracking, analytics, and consultation booking.

The system uses a user's skin profile and available product information to help identify potentially relevant skincare options. It is intended as an informational and organizational tool, not as a medical diagnostic system. Recommendations depend on the quality and completeness of the data available in the application.

2. Introduction

Skin condition can be influenced by many factors, including skin type, personal concerns, product ingredients, daily habits, hydration, and sleep. People may find it difficult to organize these details or maintain a consistent skincare routine. A digital platform can help users record relevant information and review it in one place.

This project proposes a centralized web application that supports skincare self-management. Users can create an account, maintain a profile, record lifestyle-related information, receive assessment results, explore ingredient and product information, track progress, and request a consultation with an available consultant.

3. Problem Statement

Skincare information is often spread across different products, notes, and applications. Users may struggle to connect their skin concerns with ingredient information, maintain records over time, or follow a consistent routine. A further challenge is that product choices may not account for the user's stated skin type, concerns, allergies, or sensitivities.

The project addresses these issues by providing a single platform for recording skincare-related information and presenting guidance based on the data available in the system.

4. Objectives

The main objectives of the project are:

- Provide user registration and login.
- Maintain a user skin profile containing skin type, concerns, allergies, and sensitivities.
- Support lifestyle, hydration, and sleep tracking.
- Provide a skin assessment and a skin health score based on recorded inputs.
- Generate or display structured skincare routines.
- Present ingredient information, including benefits, suitable skin types, target concerns, and warnings.
- Recommend products using the profile and product data available to the system.
- Record progress and present skincare analytics.
- Allow users to request consultations and support consultant-side status management.
- Provide administrative functions through an admin dashboard where implemented.

5. Scope of the Project

The scope includes the user-facing web application, backend API, relational database, and role-based views for users, consultants, and administrators where available. The system is designed to support informational skincare guidance, data recording, and consultation workflows.

The project does not replace a dermatologist's examination, guarantee product effectiveness, or provide emergency medical care. Any real-world use requires appropriate privacy, security, data-quality, and clinical-safety review.

6. Existing System and Proposed System

In a fragmented approach, users may track sleep and hydration separately, search for ingredients independently, keep skincare notes manually, and arrange consultations through another channel. This can make it difficult to review information in a connected way.

The proposed system centralizes these activities in one web application. It stores user-entered information, uses the skin profile and available product records for recommendation matching, and gives users a place to review assessments and progress. Consultant and admin dashboards support additional workflow management when configured.

7. Technology Stack

The project uses a web-based client-server architecture. The deployed application uses the following technologies, based on the current project setup:

- Frontend: React.js for the interactive user interface.
- Backend: Python with FastAPI for REST API endpoints.
- Database: PostgreSQL for user profiles, tracking records, assessments, products, ingredients, and consultation data.
- Database administration: pgAdmin for inspecting tables and executing SQL queries.
- Deployment: Vercel for the frontend and Render for the backend service and hosted database, as configured in the project.
- API testing: FastAPI Swagger UI at the backend /docs endpoint.

8. System Modules

The platform is divided into modules that address different parts of a user's skincare journey.

- Login and Registration: supports account creation and sign-in.
- Skin Profile: records skin type, concerns, allergies, and sensitivities.
- Lifestyle Tracking: records lifestyle habits that may relate to skin condition.
- Hydration Tracking: helps users log and review water intake.
- Sleep Tracking: records sleep information for later review.
- Skin Assessment: processes user-provided skin concern information and displays an assessment.
- Skin Health Score: provides a numerical indicator based on the application's available inputs and logic; it should not be treated as a medically validated score unless formally validated.
- Personalized Routine: presents skincare steps according to available profile information.
- Routine Generator: organizes routine steps into a structured sequence.

- Skincare Recommendations: displays guidance derived from profile and application data.
- Progress Tracking: stores dated entries so users can review changes over time.
- Skincare Analytics: summarizes recorded tracking and assessment information.
- Ingredient Assessment: displays ingredient details, potential benefits, suitable skin types, target concerns, and warnings.
- Product Recommendation: compares product information with the user's skin type and concerns and checks listed allergies or sensitivities where supported.
- Book Consultation: lets users choose an available consultant and submit a consultation request.
- Consultant Dashboard: supports viewing and updating consultation status, according to implemented permissions.
- Admin Dashboard: provides administrative views and management functions that are implemented in the application.

9. System Workflow

A typical workflow begins when a user registers or logs in. The user completes a skin profile and may record lifestyle, hydration, and sleep information. The user then submits information for skin assessment and reviews the result and skin health score. Based on available data, the application can present a skincare routine, ingredient details, and product recommendations.

The user can add progress entries and review analytics over time. If additional guidance is needed, the user can submit a consultation request. The backend stores the request, and a consultant can manage its status. Administrative functions are accessed through the admin dashboard where available.

10. Database Design Overview

The database is organized around relational tables. The current project database includes tables for users, skin profiles, skin assessments, hydration tracking, lifestyle tracking, sleep tracking, progress tracking, ingredients, and products. The consultations table stores consultation bookings and status information.

The users table provides account information and is referenced by profile, tracking, and consultation records. Skin-related records are associated with a user ID so the application can retrieve a user's information. Product and ingredient tables provide reference data for the assessment and recommendation features.

The exact column constraints and relationships should be confirmed against the deployed database schema. Production deployments should use migrations, backups, access controls, and secure handling of credentials.

11. Functional Requirements

The system should allow users to register and log in; create and update a skin profile; add tracking records; submit a skin assessment; view a skin health score; access skincare routines; view ingredient and product information; receive recommendations; record progress; review analytics; and request consultations. Consultant users should be able to manage consultation statuses where authorized. Administrators should be able to use the administrative functions implemented for their role.

12. Non-Functional Requirements

Usability: screens and forms should be understandable and consistent.

Performance: common API requests should return within a reasonable time under expected usage.

Reliability: backend errors should be logged, database transactions should be handled safely, and failed operations should return meaningful error responses.

Security and privacy: passwords and secrets must be protected; role-based authorization should prevent unauthorized access; and sensitive user information should be collected and retained only as necessary.

Maintainability: code should be modular, database changes should be documented, and endpoints should be tested after deployment.

13. Testing and Validation

Testing should include registration and login, saving and retrieving the skin profile, creating tracking entries, submitting assessments, loading ingredients, retrieving recommendations, booking consultations, and updating consultation status. API responses should be checked using Swagger UI or another API client, followed by end-to-end testing through the deployed frontend.

Successful HTTP responses alone are not sufficient proof of correctness. The returned data should also be checked for accuracy, missing values, duplicate records, and appropriate handling of invalid input. Recommendation matching should be tested with different skin types, concerns, allergies, and sensitivities.

14. Limitations

The quality of the results depends on the information entered by users and the product data maintained in the database. Rule-based matching may not capture every ingredient interaction or individual skin response. Lifestyle correlations do not establish medical causation, and the skin health score should not be interpreted as a diagnosis.

The current implementation may not include validated machine-learning models, a complete commercial product catalogue, live video consultation, or clinically verified recommendations. These capabilities should only be claimed when they have been implemented and tested.

15. Future Enhancements

Possible future improvements include validating any AI models on appropriate datasets, expanding and verifying the product catalogue, improving ingredient and allergy matching, adding reminders for routines and tracking, offering clearer charts for progress analytics, strengthening authentication and role-based access, and improving automated tests and deployment monitoring.

Further development could include secure integration with licensed healthcare providers and more robust consultation scheduling. Any health-related feature should be reviewed for privacy, safety, fairness, and appropriate medical disclaimers before real-world use.

16. Conclusion

AI Skin Intelligence and Personalized Skincare Planner aims to provide a centralized place for users to record skincare information, explore ingredient and product details, follow routines, monitor progress, and request consultations. Its modular structure combines a React frontend, FastAPI backend, and PostgreSQL database.

The project demonstrates how a web application can organize skincare-related data and present personalized informational guidance. Further testing, data validation, security improvements, and expert review can help improve its reliability and usefulness.

17. References

1. React Documentation, https://react.dev/

2. FastAPI Documentation, https://fastapi.tiangolo.com/

3. PostgreSQL Documentation, https://www.postgresql.org/docs/

4. Vercel Documentation, https://vercel.com/docs

5. Render Documentation, https://render.com/docs

6. The project's source code, database schema, API endpoints, and live application, as maintained by the project team.